CMPG 322 - Decision Support Systems (DSS) 2
SU4 - Inventory Control Models



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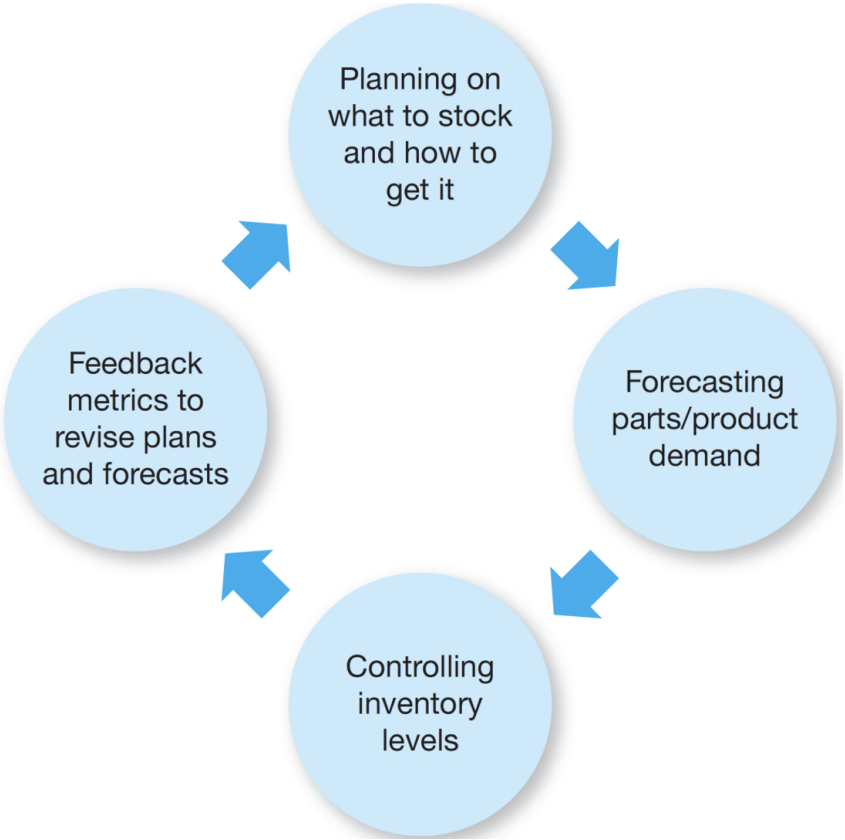
Outline

- The importance of inventory control;
- The various types of inventory related decisions;
- The economic order quantity (EOQ);
- The reorder point (ROP);
- Inventory problems that allow non-instantaneous receipt;
- Inventory problems that allow quantity discounts;
- The use of safety stock;
- Single period inventory quantities using marginal analysis;
- Dependent demand: the case for material requirements planning.

- Inventory is an expensive and important asset
- Any stored resource used to satisfy a current or future need
  - Raw materials
  - Work-in-process
  - Finished goods
- Balance high and low inventory levels to minimize costs.

- Lower inventory levels
  - Can reduce costs
  - May result in stockouts and dissatisfied customers
- All organizations have some type of inventory planning and control system
- Determine what goods/services are produced or purchased

FIGURE 6.1 Inventory Planning and Control



Five uses of inventory

- The decoupling function
- Storing resources
- Irregular supply and demand
- Quantity discounts
- Avoiding stockouts and shortages

Importance of Inventory Control

Decoupling inventory

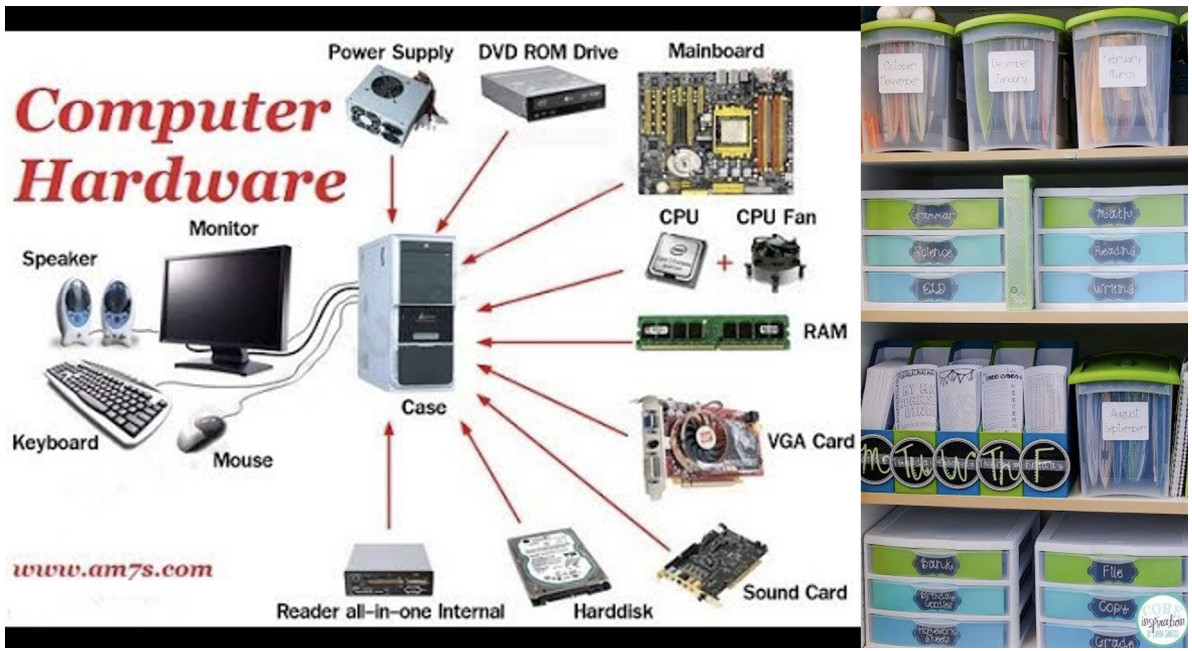
- Reduces delays and improves efficiency
- A buffer between stages



Importance of Inventory Control

Storing resources

- Seasonal products stored to satisfy off-season demand
- Materials stored as raw materials, work-in-process, or finished goods
- Labor can be stored as a component of partially completed subassemblies



Importance of Inventory Control

Irregular supply and demand

- Not constant over time
- Inventory used to buffer the variability





Importance of Inventory Control

Quantity discounts

- Lower prices may be available for larger orders
- Higher storage and holding costs
- More cash invested



Importance of Inventory Control

Avoiding stockouts and shortages

- Stockouts may result in lost sales
- Dissatisfied customers may choose to buy from another supplier
- Loss of goodwill



Inventory Cost Factors

- Two fundamental decisions
  1. How much to order
  2. When to order
- Major objective is to minimize total inventory costs
  1. Cost of the items (purchase cost or material cost)
  2. Cost of ordering
  3. Cost of carrying, or holding, inventory
  4. Cost of stockouts

TABLE 6.1 Inventory Cost Factors

ORDERING COST FACTORS	CARRYING COST FACTORS
Developing and sending purchase orders	Cost of capital
Processing and inspecting incoming inventory	Taxes
Bill paying	Insurance
Inventory inquiries	Spoilage
Utilities, phone bills, and so on for the purchasing department	Theft
Salaries and wages for purchasing department employees	Obsolescence
Supplies such as forms and paper for the purchasing department	Salaries and wages for warehouse employees
	Utilities and building costs for the warehouse
	Supplies such as forms and paper for the warehouse

- Ordering costs are generally independent of order quantity
  - Many involve personnel time
  - The amount of work is the same no matter the size of the order
- Holding costs generally vary with the amount of inventory or order size
  - Labor, space, and other costs increase with order size
  - Cost of items purchased can vary with quantity discounts

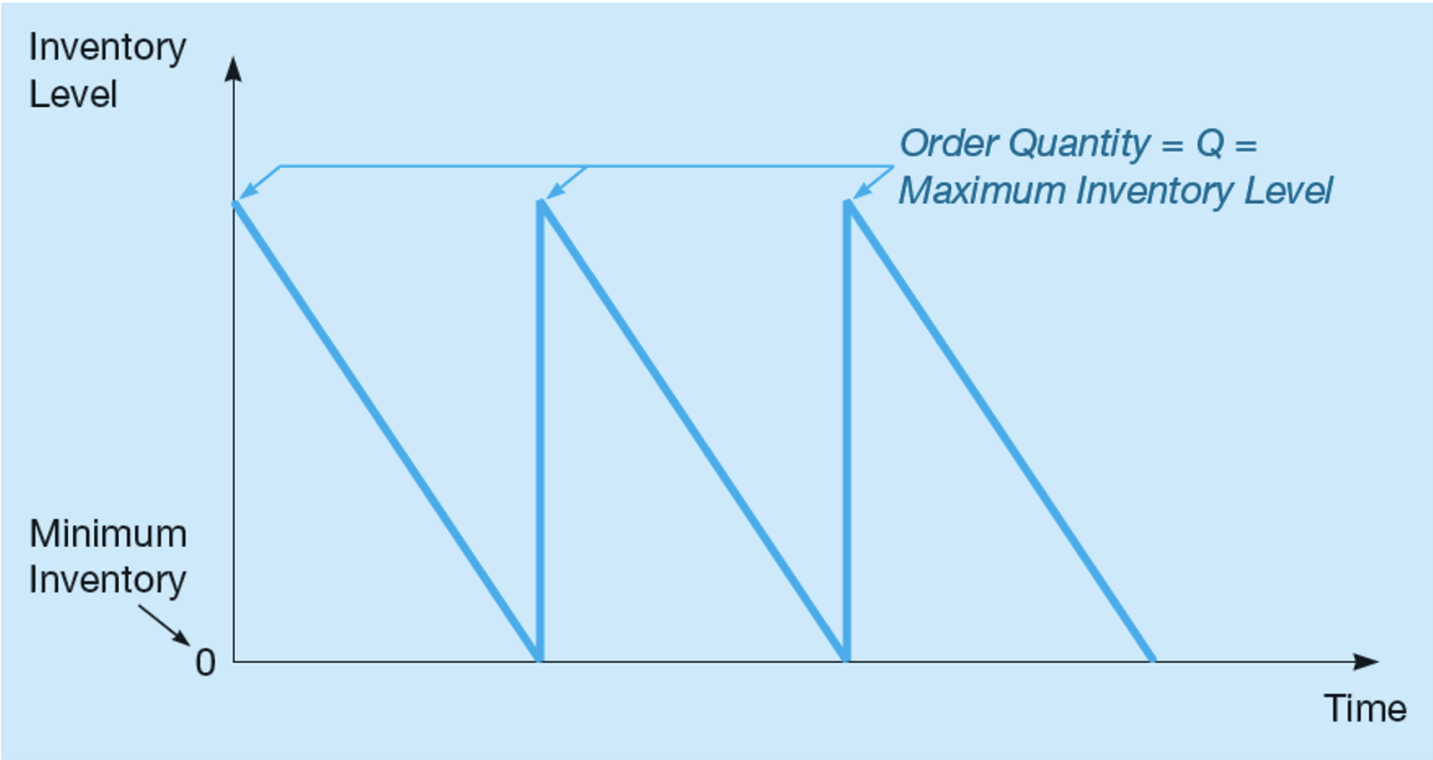
Economic Order Quantity (EOQ)

- Economic order quantity (EOQ) model
  - One of the oldest and most commonly known inventory control techniques
  - Easy to use
  - A number of important assumptions
- Objective is to minimize total cost of inventory

Assumptions:

1. Demand is known and constant over time
2. Lead time is known and constant
3. Receipt of inventory is instantaneous
4. Purchase cost per unit is constant
5. The only variable costs are ordering cost and holding or carrying cost, and these are constant throughout the year
6. Orders are placed so that stockouts or shortages are avoided completely

FIGURE 6.2 Inventory Usage over Time



Minimum Inventory

- **Annual ordering cost** is number of orders per year times cost of placing each order
- **Annual carrying cost** is the average inventory times carrying cost per unit per year

Average Inventory Level $= \frac{Q}{2}$

DAY	BEGINNING	ENDING	AVERAGE
April 1 (order received)	10	8	9
April 2	8	6	7
April 3	6	4	5
April 4	4	2	3
April 5	2	0	1

- Maximum level April 1 = 10 units
- Total of daily averages $= 9 + 7 + 5 + 3 + 1 = 25$
- Number of days = 5
- Average inventory level $= 25 \div 5 = 5$ units

Inventory costs in the EOQ

- Q = number of pieces to order
- EOQ = Q^* = optimal number of pieces to order
- D = annual demand in units for the inventory item
- C_o = ordering cost of each order
- C_h = holding or carrying cost per unit per year

Annual Ordering Cost = (Number of Orders Placed per Year) x (Ordering Cost Per Order)

$= (\frac{Annual\ Demand}{Number\ of\ Units\ in\ each\ Order}) \times (Ordering\ Cost\ Per\ Order)$

$= \frac{D}{Q}C_o$

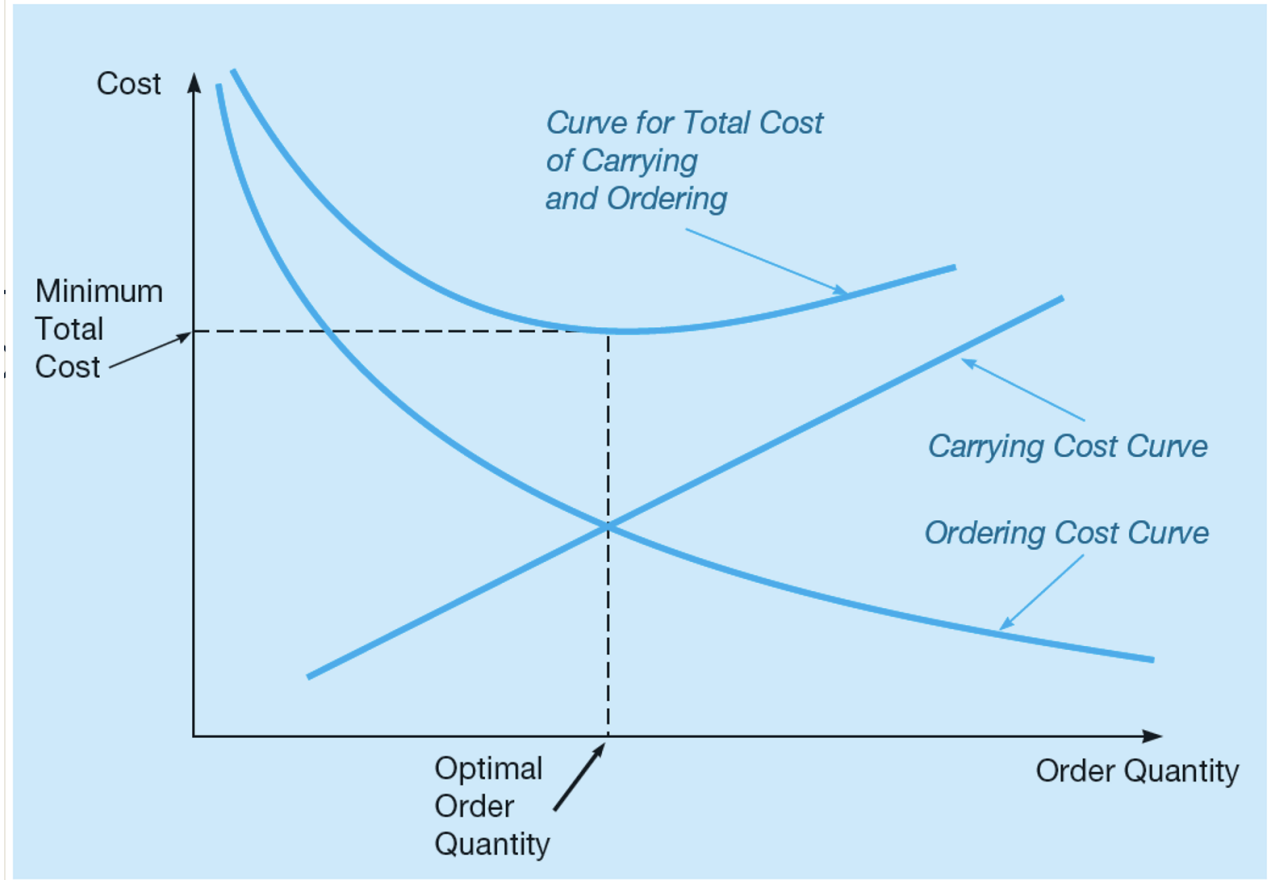
- Q = number of pieces to order
- EOQ = Q^* = optimal number of pieces to order
- D = annual demand in units for the inventory item
- C_o = ordering cost of each order
- C_h = holding or carrying cost per unit per year

Annual Holding Cost = (Average Inventory) $\times$ (Carrying Cost per Unit per Year)

$= \frac{Order\ Quantity}{2} \times$ (Carrying Cost per Unit per Year)

$= \frac{D}{Q} C_h$

FIGURE 6.3 Total Cost as a Function of Order Quantity



Finding the EOQ

- When the EOQ assumptions are met, total cost is minimized when

$$\text{Annual ordering cost} = \text{Annual holding cost}$$

$$\frac{D}{Q}C_o = \frac{Q}{2}C_h$$

Solving for Q:

$$Q^2 C_h = 2DC_o$$

$$Q^2 = \frac{2DC_o}{C_h}$$

$$Q = \sqrt{\frac{2DC_o}{C_h}}$$

Thus

$$EOQ = Q^* = \sqrt{\frac{2DC_o}{C_h}}$$

Example: Sumco Pump Company

- Sells pump housings to other companies
- Reduce inventory costs by finding optimal order quantity
  - Annual demand = 1,000 units
  - Ordering cost = \$10 per order
  - Average carrying cost per unit per year = \$0.50

$$Q^* = \sqrt{\frac{2DC_o}{C_h}}$$

- Number of orders per year: $D/Q = 5$
- Average inventory: $Q/2 = 100$ units

- Total cost:

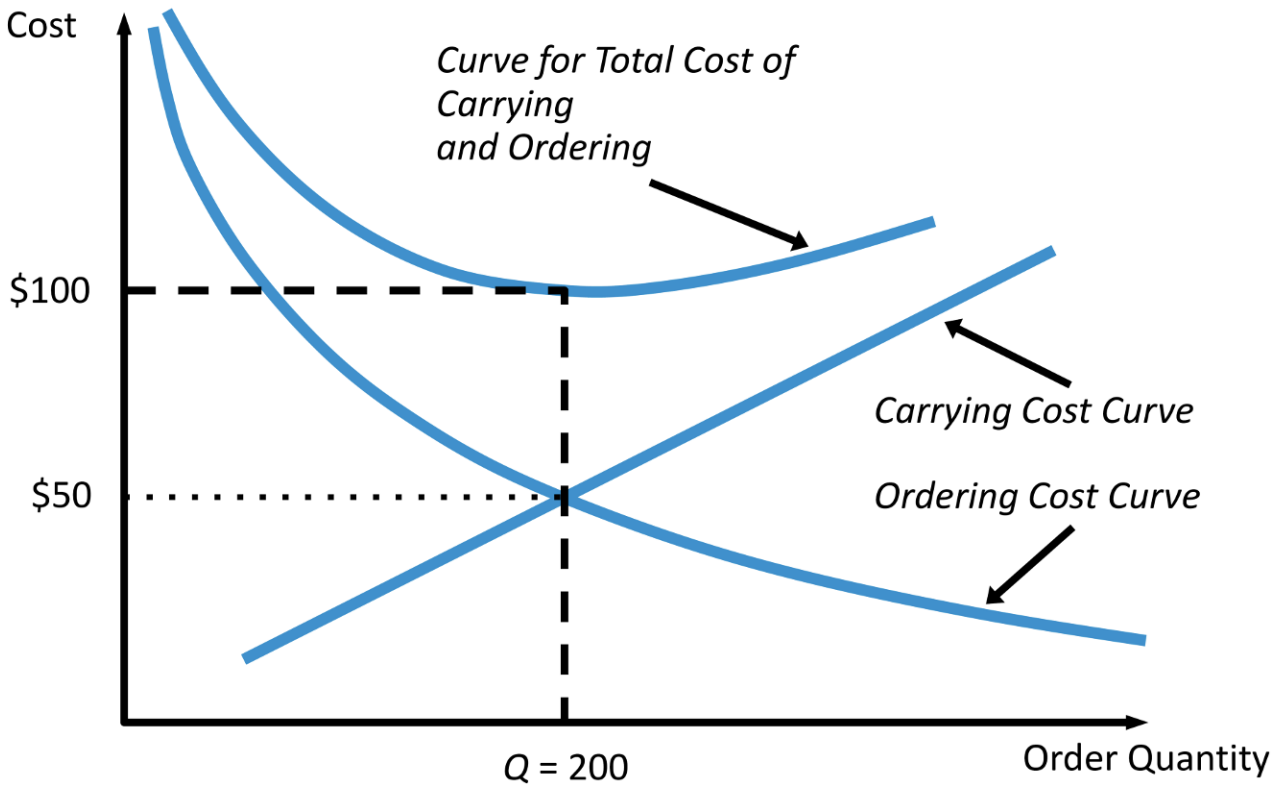
$$TC = \frac{D}{Q}C_o + \frac{Q}{2}C_h$$

$$= \frac{1,000}{200}(10) + \frac{200}{2}(0.5)$$

$$= \$50 + \$50 = \$100$$

- Number of orders per year: $D/Q = 5$
- Average inventory: $Q/2 = 100$

Example: Sumco Pump Company



- Total inventory cost can be written to include the cost of purchased items
  - *Annual purchase cost is constant at $D \times C$ no matter the order policy, where*
    - C is the purchase cost per unit
    - D is the annual demand in units
  - The average money level of inventory

$$\text{Average money level} = \frac{(CQ)}{2}$$

- Carrying costs are often expressed as an annual percentage of the unit cost or price
- Holding cost can be calculated as:

New Variable

$I =$ (Annual inventory holding charge as percentage of unit cost)

$$C_h = IC$$

$$\text{Therefore: } Q^* = \sqrt{\frac{2DC_o}{IC}}$$

- Where:
  - C_h = Cost of storing inventory for one year
  - I = Holding cost percentage
  - C = Purchase cost per unit

- The EOQ model assumes all values are known and fixed over time
- Values are estimated or may change
- **Sensitivity analysis** determines the effects of these changes
- Because the EOQ is a square root, changes in the inputs result in relatively minor changes in the order quantity

$$EOQ = Q^* = \sqrt{\frac{2DC_o}{C_h}}$$

- Sumco Pump example

$$EQC = \sqrt{\frac{2(1{,}000)(10)}{0.50}} = 200\,Units$$

- Increase C_o to \$40

$$EQC = \sqrt{\frac{2(1{,}000)(40)}{0.50}} = 400\,Units$$

- In general, the EOQ changes by the square root of the change to any of the inputs

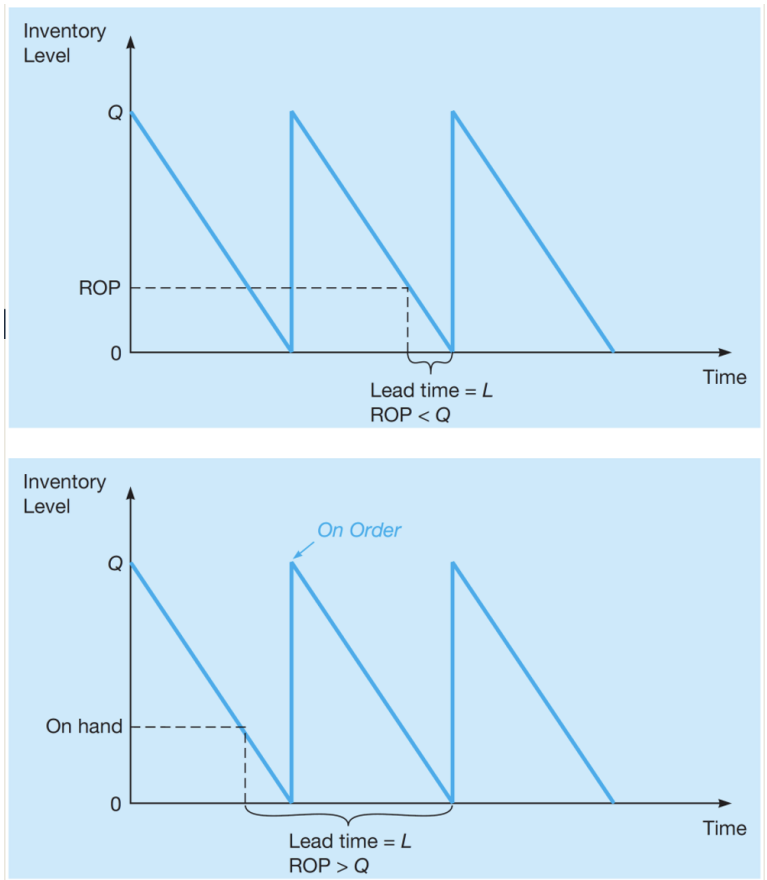
Reorder Point: When to Order

- Time between placing an order and its receipt is called the lead time (L) or delivery time.
- On hand and on order inventory must be available to meet demand during lead, the inventory position.
- Generally expressed as a reorder point (ROP).

$$\text{ROP} = (\text{Demand per day}) \times (\text{Lead time for a new order in days})$$

$$\text{ROP} = d \times L$$

FIGURE 6.4 Reorder Point Graphs



Example: Procomp Computer Chips

- Annual demand = 8,000
- Daily demand = 40 units
- Delivery in three working days

$$ROP = d \times L = 40\,units\,per\,day \times 3\,days = 120\,units$$

- An order for the EOQ (400) is placed when the inventory reaches 120 units
- The order arrives 3 days later just as the inventory is depleted

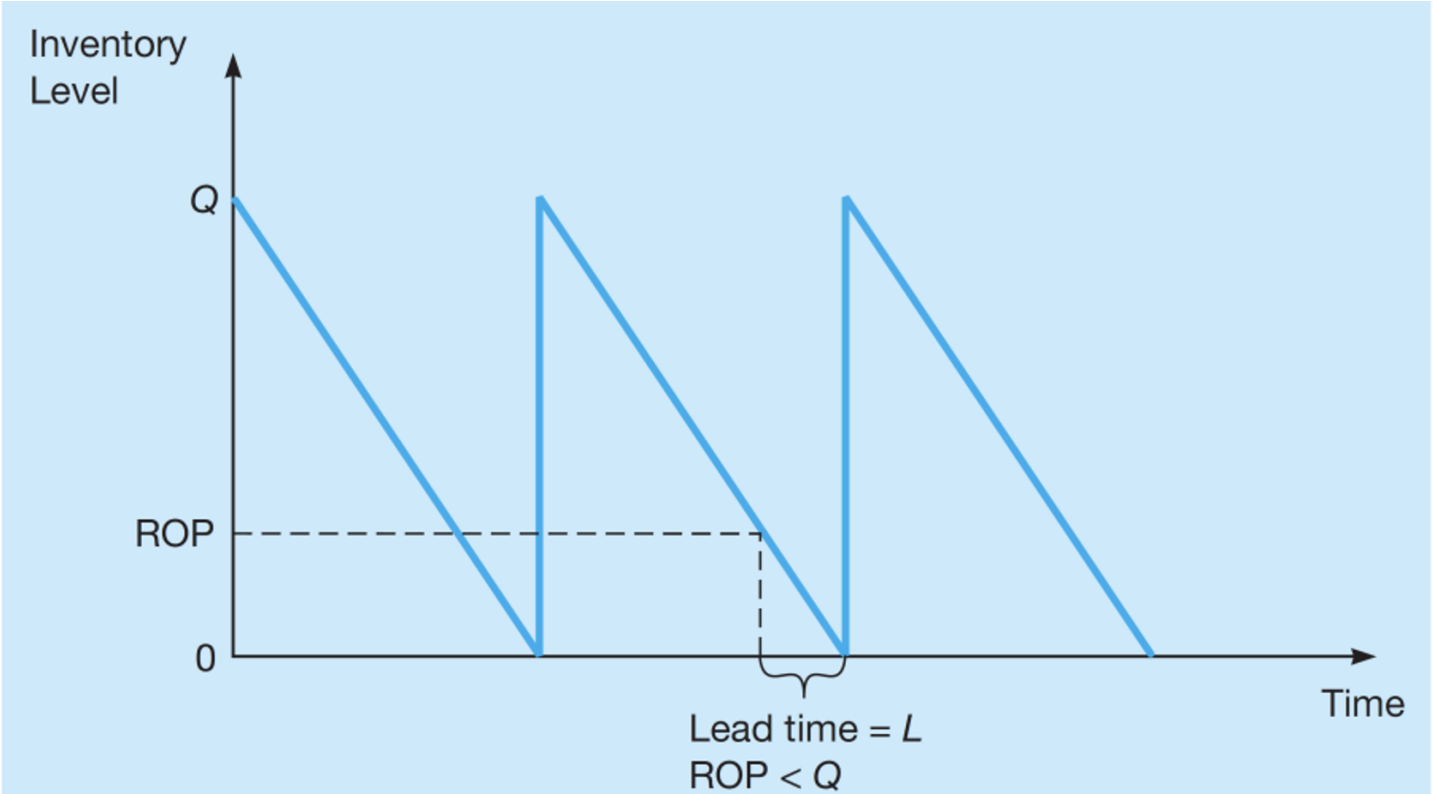
Example: Procomp Computer Chips

- Annual demand = 8,000
- Daily demand = 40 units
- Delivery in 12 working days

$$ROP = d \times L = 40\,units\,per\,day \times 12\,days = 480\,units$$

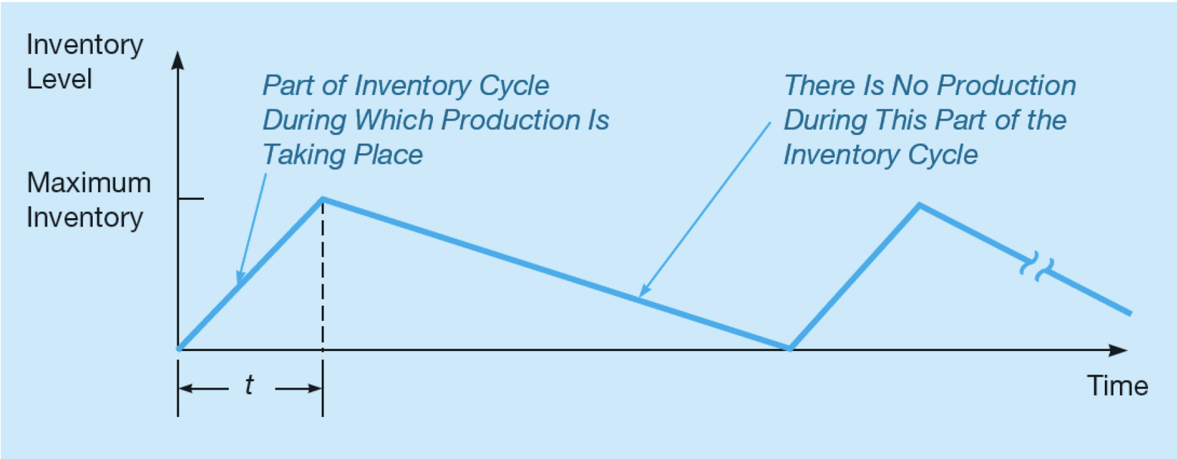
- New order placed when inventory = 80 and one order is in transit

Example: Procomp Computer Chips



- When inventory accumulates over time, the instantaneous receipt assumption does not apply
  - Daily demand rate must be taken into account
  - Production run model

FIGURE 6.5 Inventory Control and the Production Process



- Setup cost replaces ordering cost
  - Model variables
- Q = number of pieces per order, or production run
- C_s = setup cost
- C_h = holding or carrying cost per unit per year
- p = daily production rate
- d = daily demand rate
- t = length of production run in days

Annual Carrying Cost for Production Run Model

- Maximum inventory level:

$$\begin{aligned}&(\text{Total produced during the production run})\\&\quad-(\text{Total used during production run})\\&=(\text{Daily production rate})\times(\text{Number of days production})\\&\quad-(\text{Daily demand})\times(\text{Number of days production})\\&=(pt)-(dt)\end{aligned}$$

- Since $Q = pt$ and $t = \frac{Q}{p}$:

$$\text{Max. inventory} = Q - \frac{Qd}{p} = Q\left(1 - \frac{d}{p}\right)$$

Annual Carrying Cost for Production Run Model

- Average inventory is one-half the maximum:

$$\text{Average inventory} = \frac{1}{2}Q\left(1 - \frac{d}{p}\right)$$

- Annual holding cost:

$$\text{Annual holding cost} = \frac{1}{2}Q\left(1 - \frac{d}{p}\right)C_h$$

- Setup cost replaces ordering cost.
- Annual setup cost:

$$\text{Annual setup cost} = \frac{D\,C_s}{Q}$$

- (For reference) Annual ordering cost:

$$\text{Annual ordering cost} = \frac{D\,C_o}{Q}$$

Determining the Pptimal Production Quantity

- Set setup costs equal to holding costs and solve for the optimal order quantity.
- Annual holding cost = Annual setup cost.
- Solving for Q:

$$Q^* = \sqrt{\frac{2DC_s}{C_h}\left(1 - \frac{d}{p}\right)}$$

- Alternatively:

$$\frac{1}{2}Q\left(1 - \frac{d}{p}\right)C_h = \frac{DC_s}{Q}$$

Annual Costs

$$\text{Annual holding cost} = \frac{Q}{2}\left(1 - \frac{d}{p}\right)C_h$$

$$\text{Annual setup cost} = \frac{D}{Q}C_s$$

Optimal Quantity

$$\text{Optimal production quantity } Q^* = \sqrt{\frac{2DC_s}{C_h\left(1 - \frac{d}{p}\right)}}$$

Standard EOQ Comparison

$$EOQ = Q^* = \sqrt{\frac{2DC_o}{C_h}}$$

Example: Brown Manufacturing

- Produces commercial refrigeration units in batches

Given Parameters

Annual demand	D	$= 10{,}000$ units
Setup cost	C_s	$= \$100$
Carrying cost	C_h	$= \$0.50$ per unit per year
Daily production rate	p	$= 80$ units daily
Daily demand rate	d	$= 60$ units daily

1. How many units should Brown produce in each batch?
2. How long should the production part of the cycle last?

Optimal Production Quantity Calculation

$$Q^* = \sqrt{\frac{2DC_s}{C_h\left(1-\frac{d}{p}\right)}}$$

$$= \sqrt{\frac{2 \times 10,000 \times 100}{0.5\left(1-\frac{60}{80}\right)}}$$

$$= \sqrt{\frac{2,000,000}{0.5 \times 0.25}}$$

$$= \sqrt{\frac{2,000,000}{0.125}}$$

$$= \sqrt{16,000,000} = 4,000 \text{ units}$$

Production Cycle Duration

$$\text{Production time} = \frac{Q}{p} = \frac{4,000}{80} = 50 \text{ days}$$

- Basic EOQ model is adjusted by adding in the purchase or materials cost

$$\text{Total cost} = DC + \frac{D}{Q}C_o + \frac{Q}{2}C_h$$

where

D = annual demand in units
C_o = ordering cost of each order
C = cost per unit
C_h = holding or carrying cost per unit per year

- Holding cost per unit is based on cost, so:

$$C_h = IC$$

where

I = holding cost as a percentage of the unit cost (C)

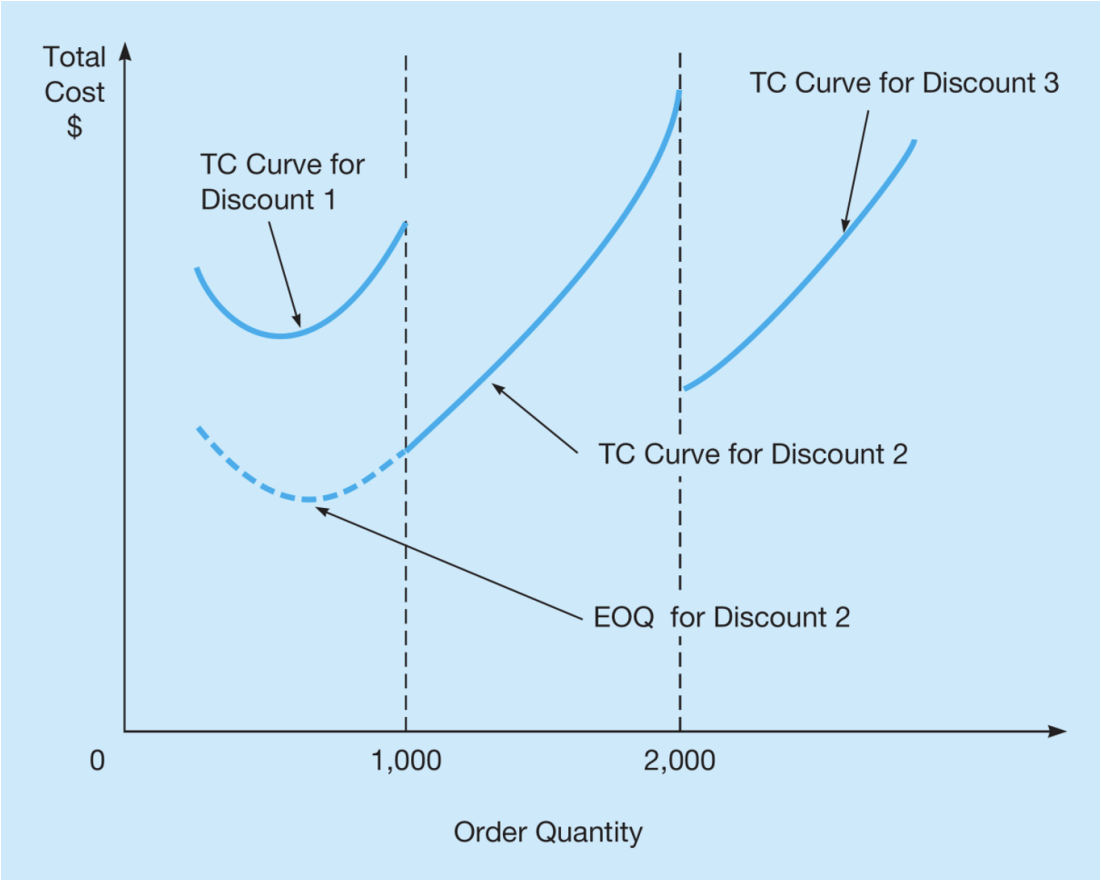
- Discount schedule and EOQs might not align
- Buying at the lowest unit cost may not result in lowest total cost

Quantity Discount Schedule

Discount Number	Discount Quantity	Discount (%)	Discount Cost ($)
1	0 to 999	0	5.00
2	1,000 to 1,999	4	4.80
3	2,000 and over	5	4.75

- Need to evaluate total cost at each price break
- Consider both purchase costs and carrying costs

FIGURE 6.6 Total Cost (TC) Curve for the Quantity Discount Model



Steps in the Process

1. For each discount price (C), compute $EOQ = \sqrt{\frac{2DC_o}{IC}}$
2. If $EOQ < Minimum$ for discount, adjust quantity to $Q = Minimum$ for discount
3. For each EOQ or adjusted Q, compute **Total cost** $= DC + \frac{D}{Q}C_o + \frac{Q}{2}C_h$
4. Choose the lowest-cost quantity

- Note: $C_h = IC$ where I is holding cost percentage
- Must evaluate all price breaks to find true minimum

Example: Brass Department Store

Discount Schedule

Disc.#	Quantity	Disc.%	Unit Cost ($)
1	0–999	0	5.00
2	1,000–1,999	4	4.80
3	2,000+	5	4.75

Step 1: Compute EOQ for Each Discount

Given: $C_o = \$49,\ D = 5,000,\ I = 0.2$

$$EOQ_1 = \sqrt{\frac{2 \times 5,000 \times 49}{0.2 \times 5.00}} = 700 \text{ units}$$

$$EOQ_2 = \sqrt{\frac{2 \times 5,000 \times 49}{0.2 \times 4.80}} = 714 \text{ units}$$

$$EOQ_3 = \sqrt{\frac{2 \times 5,000 \times 49}{0.2 \times 4.75}} = 718 \text{ units}$$

Discount Schedule

Disc.#	Quantity	Disc.%	Unit Cost ($)
1	0–999	0	5.00
2	1,000–1,999	4	4.80
3	2,000+	5	4.75

Step 2: Adjust Quantities

- $\mathbf{EOQ_1 = 700}$ (Valid for Discount 1 range: 0–999)
- $\mathbf{EOQ_2 = 714}$ (Below Discount 2 minimum: 1,000) $\rightarrow$ Adjust to $Q_2 = 1{,}000$
- $\mathbf{EOQ_3 = 718}$ (Below Discount 3 minimum: 2,000) $\rightarrow$ Adjust to $Q_3 = 2{,}000$

Adjusted Order Quantities

$$Q_1 = 700$$
$$Q_2 = 1,000$$
$$Q_3 = 2,000$$

Given Quantities

$$Q_1 = 700$$
$$Q_2 = 1,000$$
$$Q_3 = 2,000$$

Step 3: Compute Total Cost for Each Quantity

Disc.#	Unit Price ($)	Order Qty	Material Cost	Ordering Cost	Carrying Cost	Total ($)
1	5.00	700	25,000	350.00	350.00	25,700.00
2	4.80	1,000	24,000	245.00	480.00	24,725.00
3	4.75	2,000	23,750	122.50	950.00	24,822.50

Step 4: Optimal Decision

Choose Discount #2 (1,000 units) with lowest total cost of $24,725

- Safety stock helps prevent *stockouts*
- Can be implemented by adjusting the ROP:

$$ROP = (\text{Average demand during lead time}) + \text{Safety stock}$$

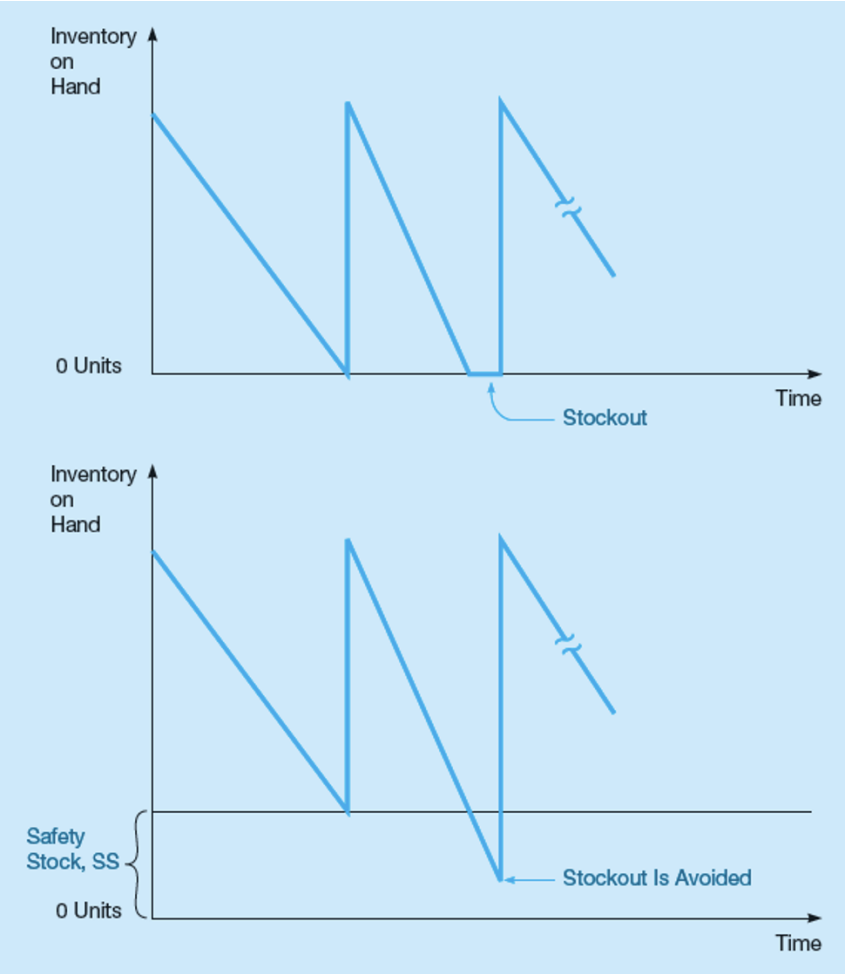
$$ROP = (\text{Average demand during lead time}) + SS$$

- Where:
  - ROP = Reorder point
  - SS = Safety stock

Key Benefit

Maintains buffer inventory to account for demand variability and supply uncertainty

FIGURE 6.7 Use of Safety Stock



- Objective is to choose a safety stock amount that minimizes total holding and stockout costs
- If variation in demand and costs are known, payoff tables could determine safety stock
- More general approach uses desired **service level** based on customer demand

Use of Safety Stock

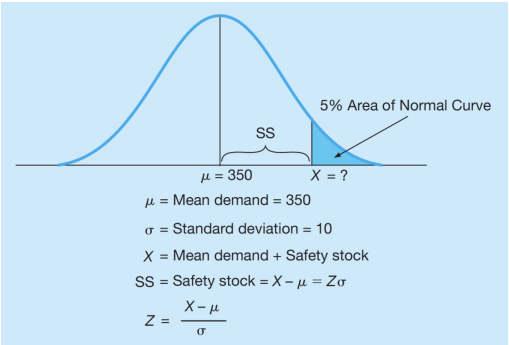
- Set safety stock to achieve a desired **service level**
  - Service level = 1 − Probability of a stockout or
  - Probability of a stockout = 1 − Service level
- ROP = (Average demand during lead time) + $Z\sigma_{LT}$ where
  - Z = number of standard deviations for a given service level
  - σ_{LT} = standard deviation of demand during the lead time
- Thus
  - Safety stock = $Z\sigma_{LT}$

For 95% service level, $Z = 1.65$ (from normal distribution table)

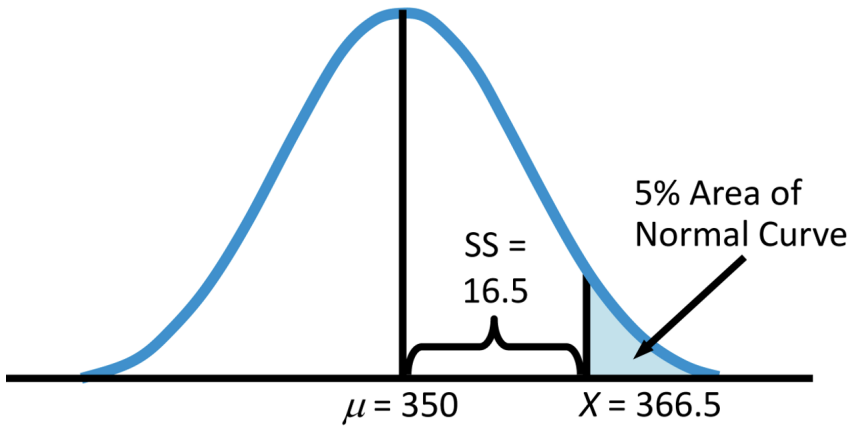
Example: Hinsdale Company

- Item A3378 has normally distributed demand during lead time
- Mean = 350 units, standard deviation = 10
- Stockouts should occur only 5
- μ = Mean demand = 350
- σ_{dLT} = Standard deviation = 10
- X = Mean demand + Safety stock
- SS = X $-\mu = Z\sigma$

FIGURE 6.8 Safety Stock and the Normal Distribution



- From Appendix A we find Z = 1.65
  ROP = (Average demand during lead time)+ ZsdLT
  = 350 + 1.65(10)
  = 350 + 16.5 = 366.5 units (or about 367 units)



$$ROP = \overline{d}L + Z(\sigma_d\sqrt{L})$$

where:

- $\overline{d}$ = average daily demand
- L = lead time in days
- σ_d = standard deviation of daily demand

Three situations to consider:

- Demand is variable but lead time is constant
- Demand is constant but lead time is variable
- Both demand and lead time are variable

General Formula for Variable Demand/Lead Time

$$ROP = \overline{d}L + Z\sqrt{L\sigma_d^2 + \overline{d}^2\sigma_L^2}$$

where σ_L = standard deviation of lead time

Example: Hinsdale Company

Case (a): Variable Demand, Constant Lead Time

- SKU F5402: $\overline{d} = 15$, $\sigma_d = 3$, $L = 4$
- Desired service level $= 97\% \Rightarrow Z = 1.88$

$$ROP = \overline{d}L + Z(\sigma_d\sqrt{L}) = 15(4) + 1.88(3\sqrt{4}) = 60 + 11.28 = 71.28$$

Case (b): Constant Demand, Variable Lead Time

- SKU B7319: $d = 25$, $L = 6$, $\sigma_L = 3$
- Desired service level $= 98\% \Rightarrow Z = 2.05$

$$ROP = dL + Z(d\sigma_L) = 25(6) + 2.05(25)(3) = 150 + 153.75 = 303.75$$

Case (c): Both Variable

- SKU F9004: $\overline{d} = 20$, $\sigma_d = 4$, $\overline{L} = 5$, $\sigma_L = 2$
- Desired service level $= 94\% \Rightarrow Z = 1.55$

$$ROP = \overline{d}L + Z\sqrt{L\sigma_d^2 + \overline{d}^2\sigma_L^2} = 100 + 1.55\sqrt{1680} = 100 + 63.53 = 163.53$$

Safety Stock for SKU A3378 at Different Service Levels

Service Level (%)	Z Value	Safety Stock (units)
90	1.28	12.8
91	1.34	13.4
92	1.41	14.1
93	1.48	14.8
94	1.55	15.5
95	1.65	16.5
96	1.75	17.5
97	1.88	18.8
98	2.05	20.5
99	2.33	23.3
99.99	3.72	37.2

Total Annual Holding Cost Formula

$$THC = \frac{Q}{2}C_h + (SS)C_h$$

where:

- THC = total annual holding cost
- Q = order quantity
- C_h = holding cost per unit per year
- SS = safety stock

- Inventory models discussed so far have assumed item demand was independent
- In many situations items demand is dependent on demand for one or more other items
- In these situations **material requirements planning (MRP)** can be employed effectively

Key Characteristics

- Used for components with *dependent demand*
- Coordinates production scheduling and inventory control
- Determines *when* to order and *how much* to order

Example Applications

Manufacturing assemblies where:

- Raw materials depend on production schedules
- Components depend on final product demand

Benefits of MRP

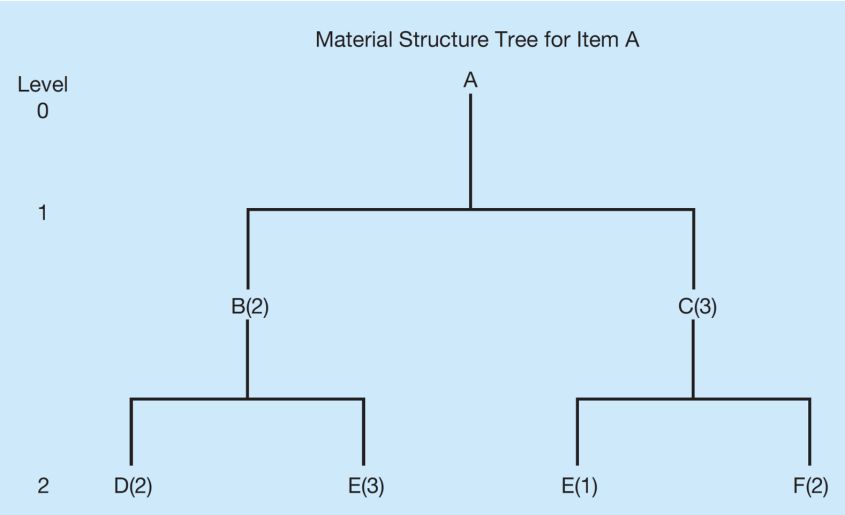
1. Increased customer service and satisfaction
2. Reduced inventory costs
3. Better inventory planning and scheduling
4. Higher total sales
5. Faster response to market changes and shifts
6. Reduced inventory levels without reduced customer service

Material Structure Tree

Material structure tree developed from the BOM

- Demand for product A is 50 units
- Each A requires 2 units of B and 3 units of C
- Each B requires 2 units of D and 3 units of E
- Each C requires 1 unit of E and 2 units of F

FIGURE 6.12 Material Structure Tree for Item A



Example Calculation

- Part B: $2\times$ number of A's $= 2 \times 50 = 100$
- Part C: $3\times$ number of A's $= 3 \times 50 = 150$
- Part D: $2\times$ number of B's $= 2 \times 100 = 200$
- Part E: $3\times$ number of B's $+ 1\times$ number of C's $= 3 \times 100 + 1 \times 150 = 450$
- Part F: $2\times$ number of C's $= 2 \times 150 = 300$

Material Structure Tree

Level 0: A
Level 1: B(2), C(3)
Level 2: D(2), E(3), E(1), F(2)

- Numbers in parentheses indicate quantity needed per parent item
- Multiple entries (like E) indicate different sources

Gross and net material requirements plan

- A gross material requirements plan is constructed after the materials structure tree is complete
  - *Time schedule*
  - Shows when an item must be ordered
  - Shows when there is no inventory on hand
  - Shows when production of an item must be started

Lead times needed for each item

- Item A - 1 week
- Item B - 2 weeks
- Item C - 1 week
- Item D - 1 week
- Item E - 2 weeks
- Item F - 3 weeks

Gross and net material requirements plan

FIGURE 6.13 Gross Material Requirements Plan for 50 Units of A

		Week 1	2	3	4	5	6	
A	Required Date						50	Lead Time = 1 Week
	Order Release					50		
Every part A requires 2 part B's in week 5								
B	Required Date					100		Lead Time = 2 Weeks
	Order Release			100				
C	Required Date					150		Lead Time = 1 Week
	Order Release				150			
D	Required Date			200				Lead Time = 1 Week
	Order Release		200					
E	Required Date			300	150			Lead Time = 2 Weeks
	Order Release	300	150					
F	Required Date				300			Lead Time = 3 Weeks
	Order Release	300						

On-Hand Inventory

Item	On-Hand Inventory
A	10
B	15
C	20
D	10
E	10
F	5

- A net material requirements plan can be constructed from:
  - The gross materials requirements plan
  - On-hand inventory information

FIGURE 6.14(a) Net Material Requirements Plan for 50 Units of A

		Week						
Item		1	2	3	4	5	6	Lead Time
A	Gross						50	1 Week
	On-Hand 10						10	
	Net						40	
	Order Receipt						40	
	Order Release					40		
B	Gross					80[A]		2 Weeks
	On-Hand 15					15		
	Net					65		
	Order Receipt					65		
	Order Release			65				

Gross and net material requirements

FIGURE 6.14
Net Material Requirements Plan for 50 Units of A

Item		Week 1	2	3	4	5	6	Lead Time
A	Gross						50	1 Week
	On-Hand 10						10	
	Net						40	
	Order Receipt						40	
	Order Release					40		
B	Gross					80[A]		2 Weeks
	On-Hand 15					15		
	Net					65		
	Order Receipt					65		
	Order Release			65				
C	Gross					120[A]		1 Week
	On-Hand 20					20		
	Net					100		
	Order Receipt					100		
	Order Release				100			
D	Gross			130[B]				1 Week
	On-Hand 10			10				
	Net			120				
	Order Receipt			120				
	Order Release		120					
E	Gross			195[B]	100[C]			2 Weeks
	On-Hand 10			10	0			
	Net			185	100			
	Order Receipt			185	100			
	Order Release	185	100					
F	Gross				200[C]			3 Weeks
	On-Hand 5				5			
	Net				195			
	Order Receipt				195			
	Order Release	195						

Questions

